# India Agricultural Productivity Intelligence Yield & Performance Analysis (1997–2019)

Data Analytics Capstone | Agriculture Sector

**Section A, Group G-17**

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# Why Agricultural Productivity Needs Data Intelligence

## Agriculture's Economic Significance

- ~18% of GDP contribution
- Employment for ~50% of workforce
- Food security for 1.4B population

## Core Problem

- No unified system for yield efficiency tracking
- Poor district-level visibility
- Leads to suboptimal policy decisions

# From Raw Data to Insights

1

## DATA SOURCE

**3.45L rows** • 23 years • 707 districts

2

## ETL PIPELINE

**Python-based** cleaning & transformation

3

## DATA QUALITY

Null handling • Outliers • Feature engineering

4

## DASHBOARDS

**Tableau dashboards** for insights

**Scalable pipeline transforming raw agricultural data into decision-ready insights**

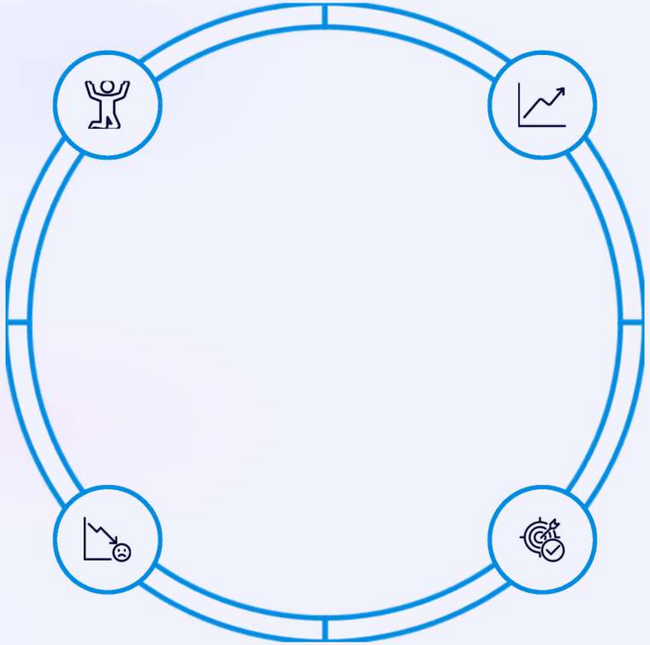
# Core Metrics Driving Agricultural Productivity

### Yield Efficiency

Output per hectare

### YoY Growth

Annual production change



### Production Loss

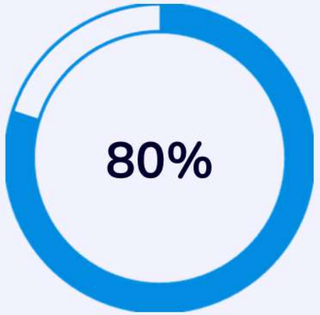
Below median yield

### Underperformance Index

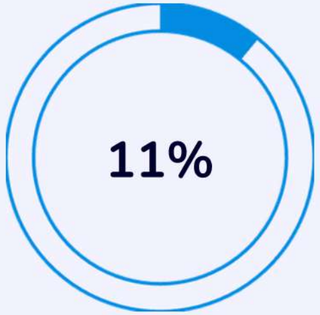
Estimated output gap

**Yield efficiency is the primary driver of agricultural productivity**

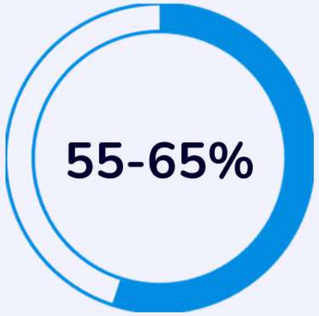
# Key Insights from EDA



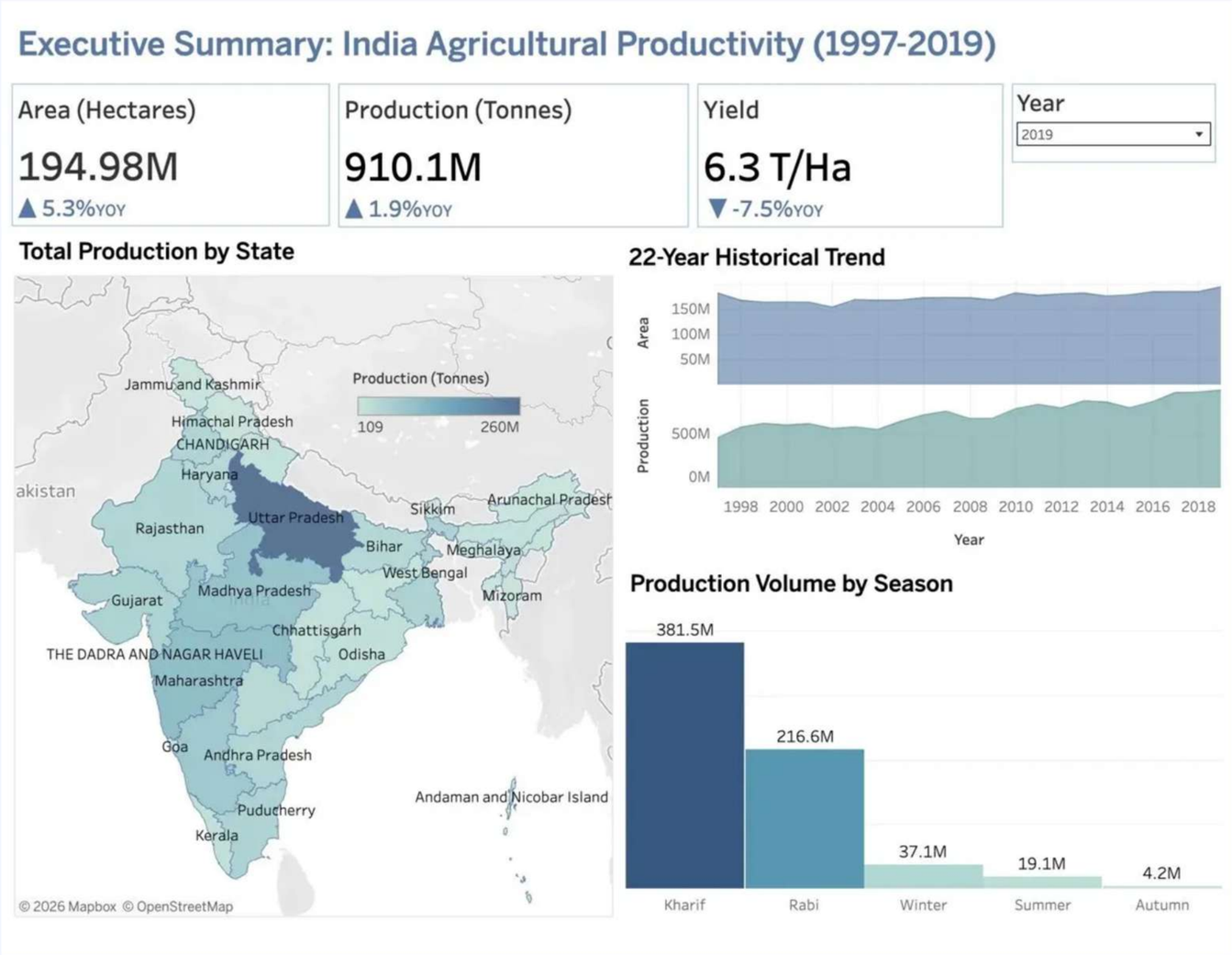

Production Growth




Area Expansion




Kharif Share



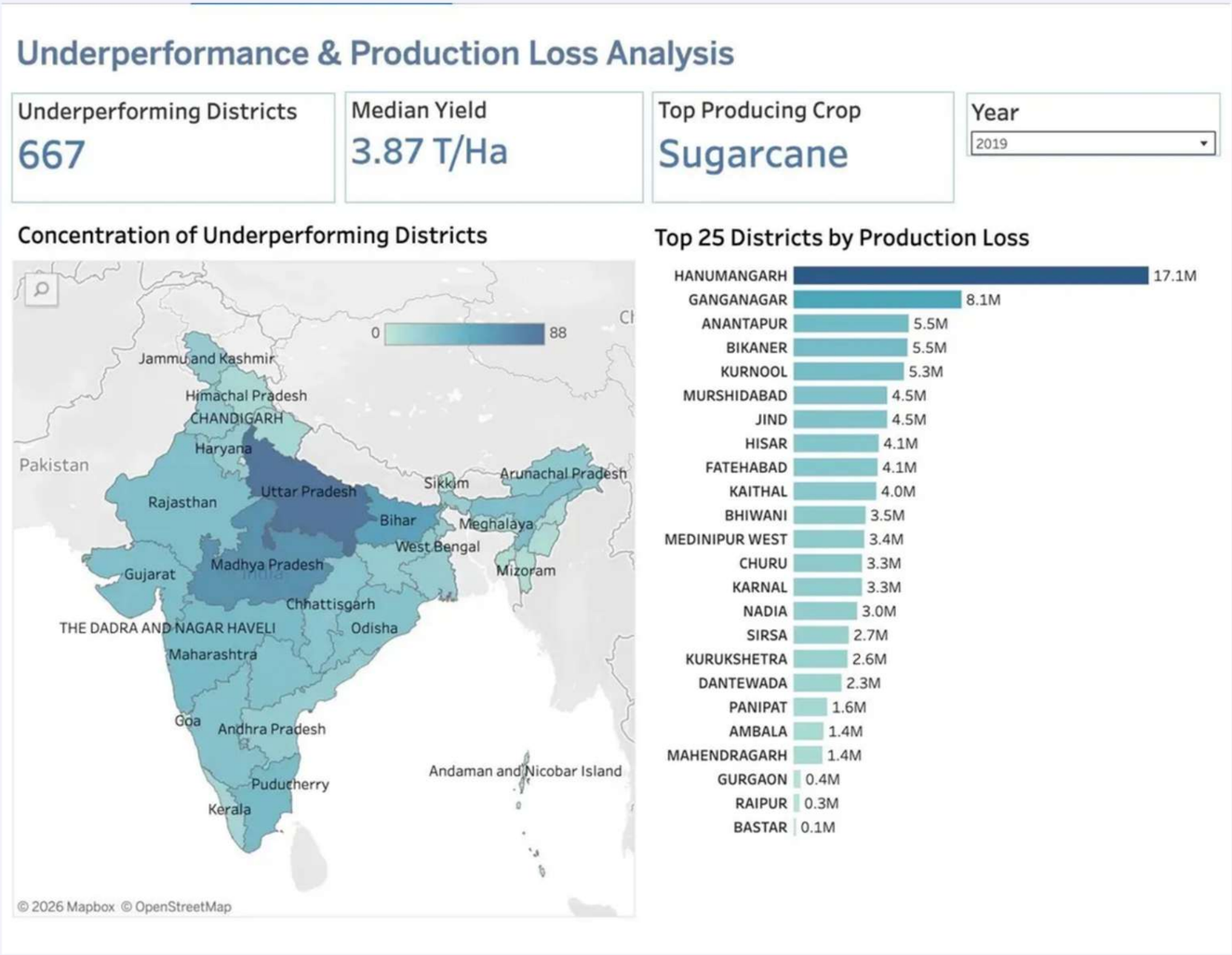

**Growth is driven primarily by yield improvements, not land expansion.**

# 99.7% of districts fall below median yield

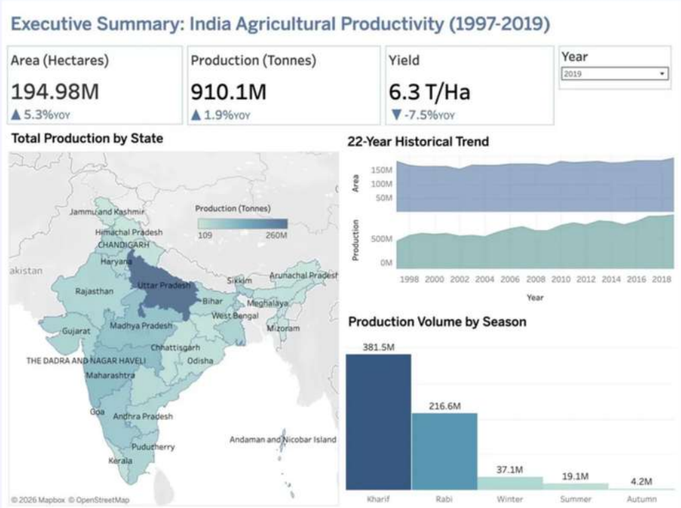
Districts underperforming median yield

Median yield (Tonnes/Hectare)(Low Benchmark)

**Most districts produce, but inefficiently →**
**Massive hidden yield loss**

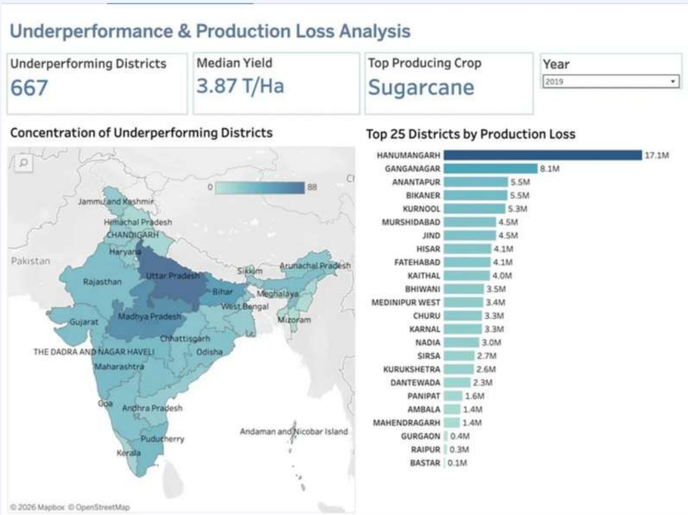


# Dashboard Walkthrough



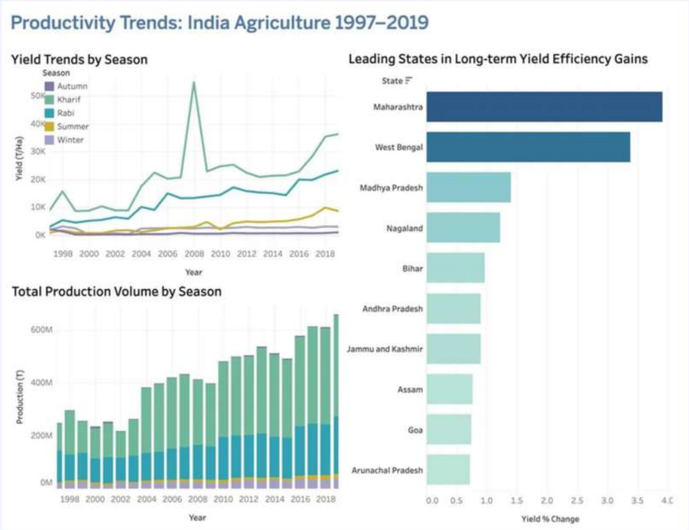

## Executive KPI View

High-level metrics and trend summaries



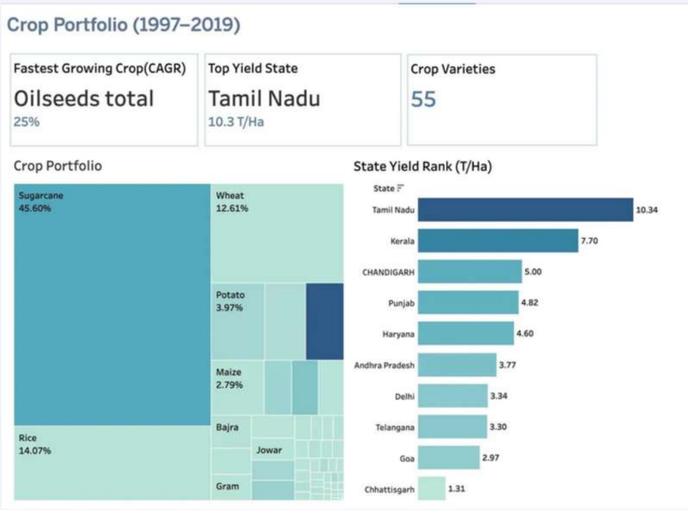

## Underperformance View

Loss quantification and gap analysis




## Trends View

Temporal analysis across years




## Crop Portfolio View

Comparative performance across commodities

# Strategic Recommendations

**Recommendations driven by yield gaps, crop patterns, and regional inefficiencies**

### Precision Intervention

Target **high-loss** districts (**identified in EDA: Central India cluster**) in Central India with customized agronomic packages and extension services

### Crop Diversification

Reduce sugarcane dependency **(top producing crop identified in dataset)**; promote drought-resistant alternatives

### Season-Based Credit

Align credit allocation cycles with Kharif/Rabi seasons to optimize working capital

### High-Efficiency Crops

Scale coconut and maize cultivation—demonstrated yield efficiency in analysis

# Impact & Value Creation

### Yield Improvement

5–10% efficiency gain = translating into large-scale production gains across districts

### Annual Impact

$2–4B annual value unlock from closing yield gaps

### Policy Targeting

Target subsidies to high-loss districts (**Central India cluster identified**)

### Implementation Feasibility

Built on existing district-level data and scalable ETL pipeline for rapid deployment

# Limitations & Future Scope

## Current Limitations

- **No post-2019 data** → limits recent trend validation
- **Missing climate variables** → limits causal analysis
- **No input cost data** → prevents profitability estimation
- **No real-time monitoring** → delayed policy response

## Future Enhancements

- ML-based yield prediction (district-level forecasting)
- Integrate satellite + weather data for real-time monitoring
- Add cost & price data for profitability insights
- Deploy real-time dashboard with IoT-enabled field data

**Next Phase**: Build a real-time decision support system for district-level agricultural optimization